

■ .NET 8 AI Migration Assistant Report

Project:	demo_project
Path:	C:\Users\grith\Desktop\Infinite\demo_project
Generated at:	2026-06-10 09:12:44
Overall Risk Rating:	Low (14/100)
Migration Readiness:	86%
Estimated Effort:	1-3 Days (Low Effort)

1. Executive Summary

This migration report evaluates the structure, dependencies, and code configuration of the application located at C:\Users\grith\Desktop\Infinite\demo_project. The assistant scanned legacy codebases, computed risk scores, and generated remediation strategies using Gemini models.

2. Migration Roadmap

- **Phase 1: Project Preparation** - Create full system backups and code checkpoints.
- **Phase 2: Target Upgrades** - Update build dependencies and change TARGET framework to net8.0.
- **Phase 3: Package Refactoring** - Upgrade legacy assemblies and NuGet dependencies.
- **Phase 4: API Remappings** - Code refactoring of WebForms/WCF bindings to Minimal APIs and gRPC.

3. Risk Blocker Summary

• **[Medium] System.Web**

Remediation: System.Web is deprecated. Replace HttpContext.Current usage with dependency injection of IHttpContextAccessor. Migrate handlers/modules to ASP.NET Core Middleware.

• **[Medium] Legacy Packages**

Remediation: Upgrade legacy NuGet packages to modern versions targeting .NET 8. Legacy packages found: EntityFramework, log4net, Microsoft.AspNet.Mvc, Microsoft.AspNet.Razor, Microsoft.AspNet.WebApi, Microsoft.AspNet.WebPages, Microsoft.AspNet.Web.Optimization, AutoFac.Mvc5, AjaxControlToolkit

• **[Low] Config Complexity**

Remediation: Convert XML configurations (Web.config/App.config) to JSON-based appsettings.json. Migrate WCF endpoint configurations to code-based initialization.

4. File-by-File AI Suggestions

File: Controllers\HomeController.cs

Summary: Legacy ASP.NET MVC controller or HTTP module 'HomeController.cs' using HttpContext.Current, Session, Application state, Cache API, and ConfigurationManager.

Strategy: 1. Replace HttpContext.Current with constructor-injected IHttpContextAccessor. 2. Replace System.Web.Mvc.Controller with Microsoft.AspNetCore.Mvc.Controller. 3. Replace ConfigurationManager with IConfiguration. 4. Replace ASP.NET Cache with IMemoryCache or IDistributedCache. 5. Replace Session direct access with HttpContext.Session extension methods. 6. Replace Application state with a singleton service registered in DI.

Migrated Equivalent Preview:

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Http;
using Microsoft.Extensions.Caching.Memory;
using Microsoft.Extensions.Configuration;
using Microsoft.EntityFrameworkCore;

namespace LegacyApp.Controllers;

/// <summary>Home controller migrated to ASP.NET Core with dependency injection.</summary>
public class HomeController : Controller
{
    private readonly IHttpContextAccessor _httpContextAccessor;
    private readonly IMemoryCache _cache;
    private readonly IConfiguration _configuration;
    private readonly AppDbContext _db;

    public HomeController(
        IHttpContextAccessor httpContextAccessor,
        IMemoryCache cache,
        IConfiguration configuration,
        AppDbContext db)
    {
        _httpContextAccessor = httpContextAccessor;
        _cache = cac
    }
    //... [code truncated for length in PDF] ...
```